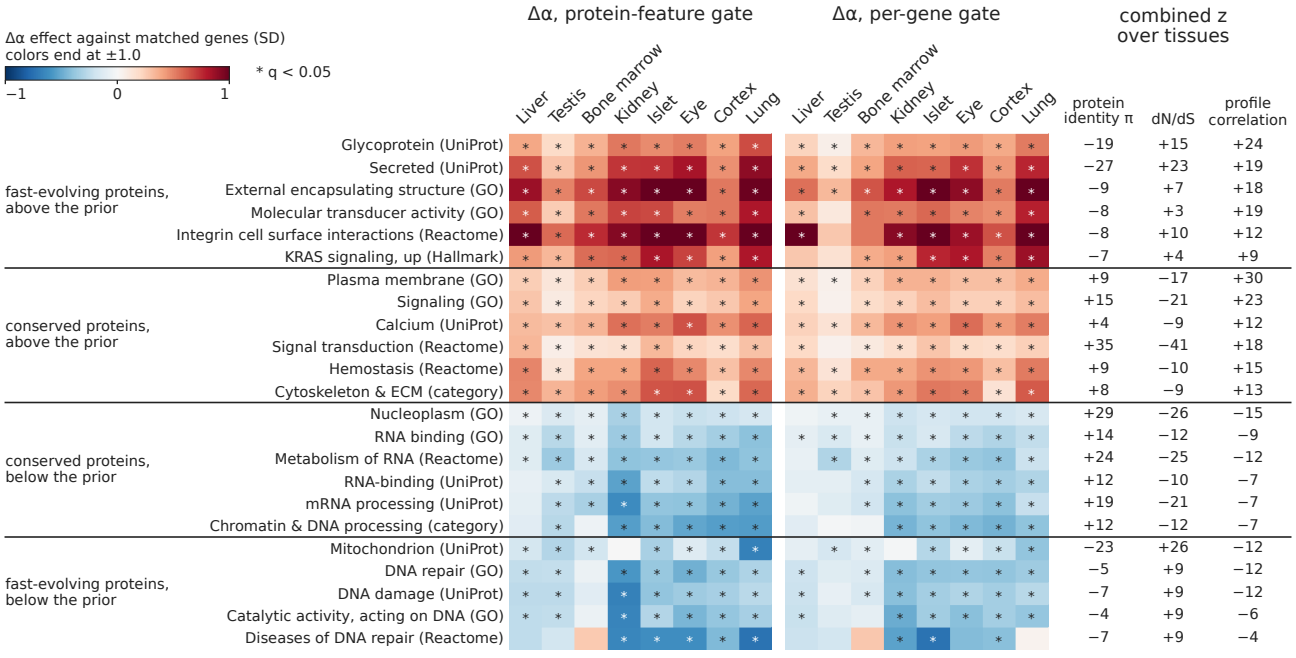
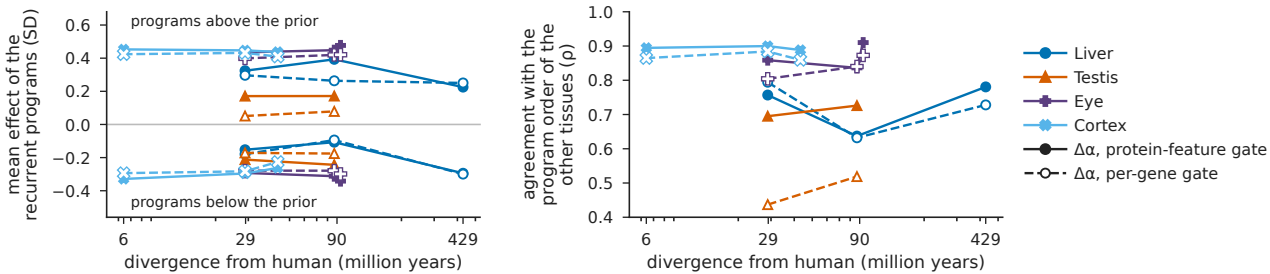

Δα effect against matched genes (SD)
colors end at ±1.0
−1
0
1
* q < 0.05
Δα, protein-feature gate
Δα, per-gene gate
combined z over tissues
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
protein identity π
dN/dS
profile correlation
fast-evolving proteins, above the prior
Glycoprotein (UniProt) −19 +15 +24
Secreted (UniProt) −27 +23 +19
External encapsulating structure (GO) −9 +7 +18
Molecular transducer activity (GO) −8 +3 +19
Integrin cell surface interactions (Reactome) −8 +10 +12
KRAS signaling, up (Hallmark) −7 +4 +9
conserved proteins, above the prior
Plasma membrane (GO) +9 −17 +30
Signaling (GO) +15 −21 +23
Calcium (UniProt) +4 −9 +12
Signal transduction (Reactome) +35 −41 +18
Hemostasis (Reactome) +9 −10 +15
Cytoskeleton & ECM (category) +8 −9 +13
conserved proteins, below the prior
Nucleoplasm (GO) +29 −26 −15
RNA binding (GO) +14 −12 −9
Metabolism of RNA (Reactome) +24 −25 −12
RNA-binding (UniProt) +12 −10 −7
mRNA processing (UniProt) +19 −21 −7
Chromatin & DNA processing (category) +12 −12 −7
fast-evolving proteins, below the prior
Mitochondrion (UniProt) −23 +26 −12
DNA repair (GO) −5 +9 −12
DNA damage (UniProt) −7 +9 −12
Catalytic activity, acting on DNA (GO) −4 +9 −6
Diseases of DNA repair (Reactome) −7 +9 −4

**(a)**


programs above the prior
programs below the prior
mean effect of the recurrent programs (SD)
0.6
0.4
0.2
0.0
−0.2
−0.4
6
29
90
429
divergence from human (million years)
agreement with the program order of the other tissues (ρ)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
divergence from human (million years)
Liver
Testis
Eye
Cortex
Δα, protein-feature gate
Δα, per-gene gate

**(b)**